Data Wrangling Homework

In the spirit of warming up for final projects, this weeks homework will be somewhat open-ended. Actually, it will more like closed-ended and open-middled...

Different doctors with different backgrounds, trained in different places, etc., might behave differently. In other words, one doctor might have various biases relative to another. We hope not, at least in critical situations, but doctors are people too.

Your job is to determine whether the 4 doctors in our data set are behaving essentially the same with respect to measuring clump thickness, bland chromatin, and diagnosis of tumor type, or whether any one of them seems to be different.

The submission should be a pdf that makes your case as though to a boss/hospital administrator; it should make the case in enough detail to be convincing, but not in such detail that your boss will hate you. For example, one doctor-to-doctor comparison can be described in some detail, but the rest can be summarized with "Similar comparisons were made for ..."

Your final conclusion should be whether 1) everything seems okay with respect to the doctors or 2) if there are any red flags that might warrent further scrutiny.

Do the analysis with an open mind. It's not good to enter an analysis with a pre-conceived notion of what you may or may not find.

In [6]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
def bcd_load_clean():
    bcd = pd.read_csv('./data/breast_cancer_data.csv')
    bcd['patient_id'] = bcd['patient_id'].astype('string')
    bcd['doctor_name'] = bcd['doctor_name'].str.split().str[1]
    bcd['bare_nuclei'] = bcd['bare_nuclei'].replace('?', '')
    bcd['bare_nuclei'] = pd.to_numeric(bcd['bare_nuclei'])

    return bcd
```

In [3]:
```
bcd = bcd_load_clean()
```

In [4]:
```
bcd2 = bcd.drop(labels = ['patient_id', 'cell_size_uniformity', 'cell_shape_
                          'cell_shape_uniformity', 'marginal_adhesion', 'sir
                          'bare_nuclei', 'normal_nucleoli', 'mitoses'],
                axis = 1,             # we're selecting column - default is
                inplace = False)      # we could modify bcd itself with True
```

In [8]:
```
bcd2
```

Out[8]:

	clump_thickness	bland_chromatin	class	doctor_name
0	5.0	3.0	benign	Doe
1	5.0	3.0	benign	Smith
2	3.0	3.0	benign	Lee
3	6.0	3.0	benign	Smith
4	4.0	3.0	benign	Wong
...	...	...	...	...
694	3.0	1.0	benign	Lee
695	2.0	1.0	benign	Smith
696	5.0	8.0	malignant	Lee
697	4.0	10.0	malignant	Lee
698	4.0	10.0	malignant	Wong

699 rows × 4 columns

In [7]:
```
bcd2.info()
bcd2.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 699 entries, 0 to 698
Data columns (total 4 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   clump_thickness  698 non-null    float64
 1   bland_chromatin  695 non-null    float64
 2   class            699 non-null    object
 3   doctor_name      699 non-null    object
dtypes: float64(2), object(2)
memory usage: 22.0+ KB
```

Out[7]:

	clump_thickness	bland_chromatin
count	698.000000	695.000000
mean	4.416905	3.447482
std	2.817673	2.441191
min	1.000000	1.000000
25%	2.000000	2.000000
50%	4.000000	3.000000
75%	6.000000	5.000000
max	10.000000	10.000000

Summaries of doctors

In [13]:

```
bcd2['doctor_name'] = bcd2['doctor_name'].astype('category')

doctors = bcd2['doctor_name'].unique()

# Descriptive stats per doctor
for feature in ['clump_thickness', 'bland_chromatin']:
    print(f"\n {feature} by Doctor")
    display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.mi
```

```
clump_thickness by Doctor
```

```
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to retain
current behavior or observed=True to adopt the future default and silence th
is warning.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function mean at 0x104b21440> is
currently using SeriesGroupBy.mean. In a future version of pandas, the provi
ded callable will be used directly. To keep current behavior pass the string
"mean" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function std at 0x104b21580> is c
urrently using SeriesGroupBy.std. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"std" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function min at 0x104b20b80> is c
urrently using SeriesGroupBy.min. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"min" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function max at 0x104b20a40> is c
urrently using SeriesGroupBy.max. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"max" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
```

doctor_name	mean	std	min	max
Doe	4.189189	3.043506	1.0	10.0
Lee	4.182320	2.504422	1.0	10.0
Smith	4.874286	2.851934	1.0	10.0
Wong	4.445860	2.806574	1.0	10.0

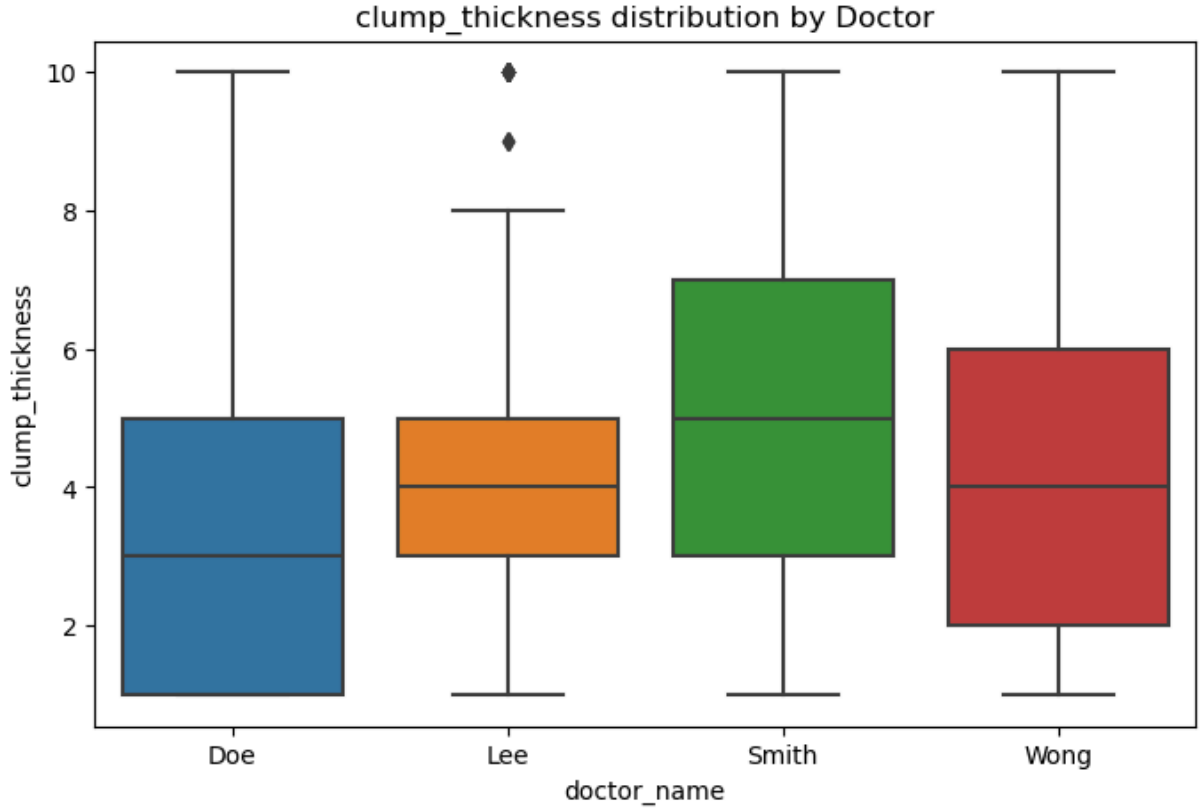
```
bland_chromatin by Doctor
```

```
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to retain
current behavior or observed=True to adopt the future default and silence th
is warning.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function mean at 0x104b21440> is
currently using SeriesGroupBy.mean. In a future version of pandas, the provi
ded callable will be used directly. To keep current behavior pass the string
"mean" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function std at 0x104b21580> is c
urrently using SeriesGroupBy.std. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"std" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function min at 0x104b20b80> is c
urrently using SeriesGroupBy.min. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"min" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/322622537.p
y:8: FutureWarning: The provided callable <function max at 0x104b20a40> is c
urrently using SeriesGroupBy.max. In a future version of pandas, the provide
d callable will be used directly. To keep current behavior pass the string
"max" instead.
  display(bcd2.groupby('doctor_name')[feature].agg([np.mean, np.std, np.min,
np.max]))
```

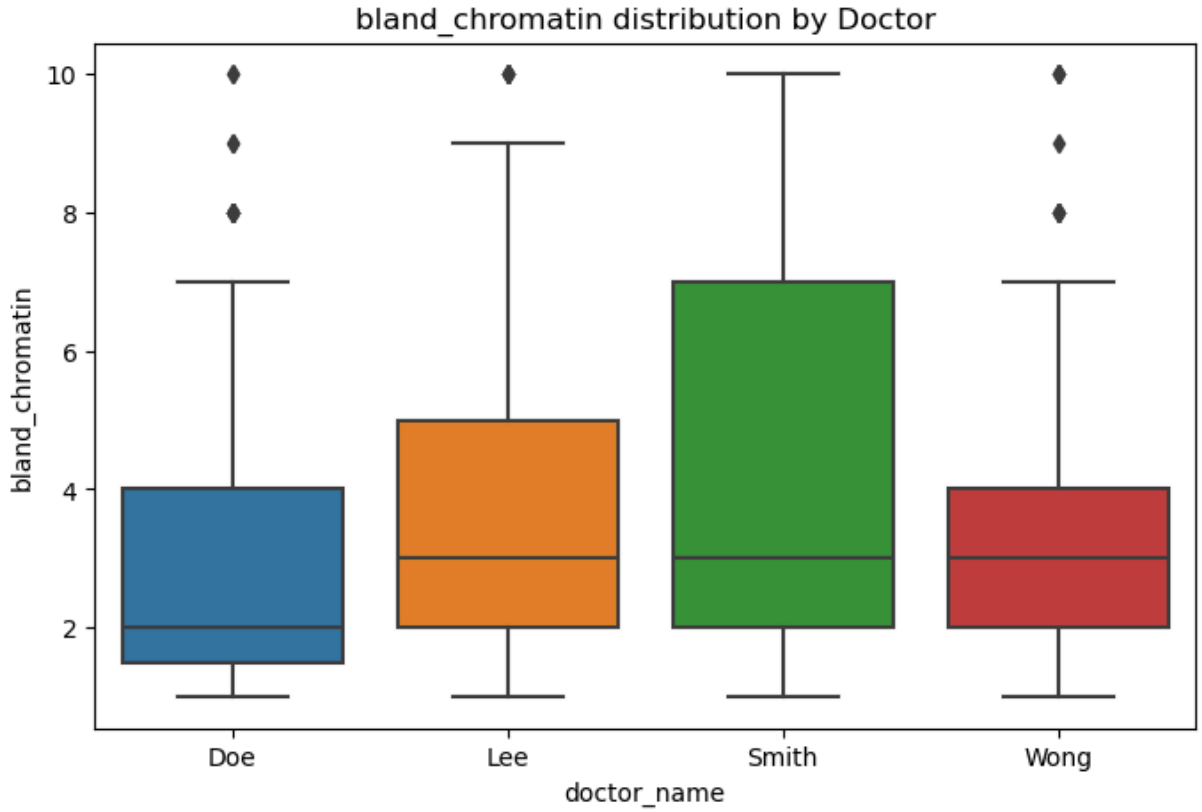
doctor_name	mean	std	min	max
Doe	3.076503	2.200061	1.0	10.0
Lee	3.435754	2.431085	1.0	10.0
Smith	3.863636	2.774092	1.0	10.0
Wong	3.426752	2.265131	1.0	10.0

In [12]:
```
for feature in ['clump_thickness', 'bland_chromatin']:
    plt.figure(figsize=(8,5))
    sns.boxplot(x='doctor_name', y=feature, data=bcd2)
    plt.title(f"{feature} distribution by Doctor")
    plt.show()
```

```
/opt/anaconda3/lib/python3.11/site-packages/seaborn/categorical.py:641: Futu
reWarning: The default of observed=False is deprecated and will be changed t
o True in a future version of pandas. Pass observed=False to retain current
behavior or observed=True to adopt the future default and silence this warni
ng.
  grouped_vals = vals.groupby(grouper)
```




```
/opt/anaconda3/lib/python3.11/site-packages/seaborn/categorical.py:641: Futu
reWarning: The default of observed=False is deprecated and will be changed t
o True in a future version of pandas. Pass observed=False to retain current
behavior or observed=True to adopt the future default and silence this warni
ng.
  grouped_vals = vals.groupby(grouper)
```



Tumor type distribution per doctor

In [14]:
```
display(bcd2.groupby('doctor_name')['class'].value_counts(normalize=True).ur
```

```
/var/folders/71/br_33dt53n948nsmj9czsvr80000gn/T/ipykernel_61316/3754176847.
py:1: FutureWarning: The default of observed=False is deprecated and will be
changed to True in a future version of pandas. Pass observed=False to retain
current behavior or observed=True to adopt the future default and silence th
is warning.
  display(bcd2.groupby('doctor_name')['class'].value_counts(normalize=True).
unstack())
```

class	benign	malignant
doctor_name		
Doe	0.686486	0.313514
Lee	0.668508	0.331492
Smith	0.579545	0.420455
Wong	0.687898	0.312102

While Drs Doe, Lee, and Wong seem to have near identical values for most of these analyses, Dr. Smith tends to attribute higher clump thickness and bland chromatin to each tumor, leading to a much higher prevalence in malignant tumors in his reporting compared to the other three doctors.